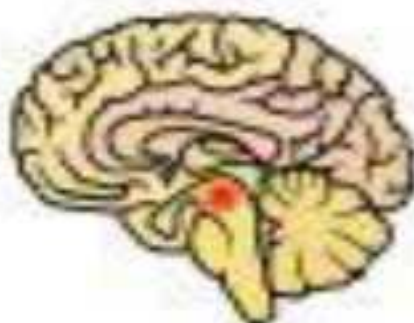




Parkinson's Disease

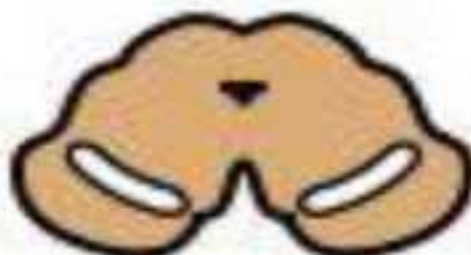
tremor



dopamine



**Diminished
substantia nigra**



**Neuron affected
by Parkinson's**



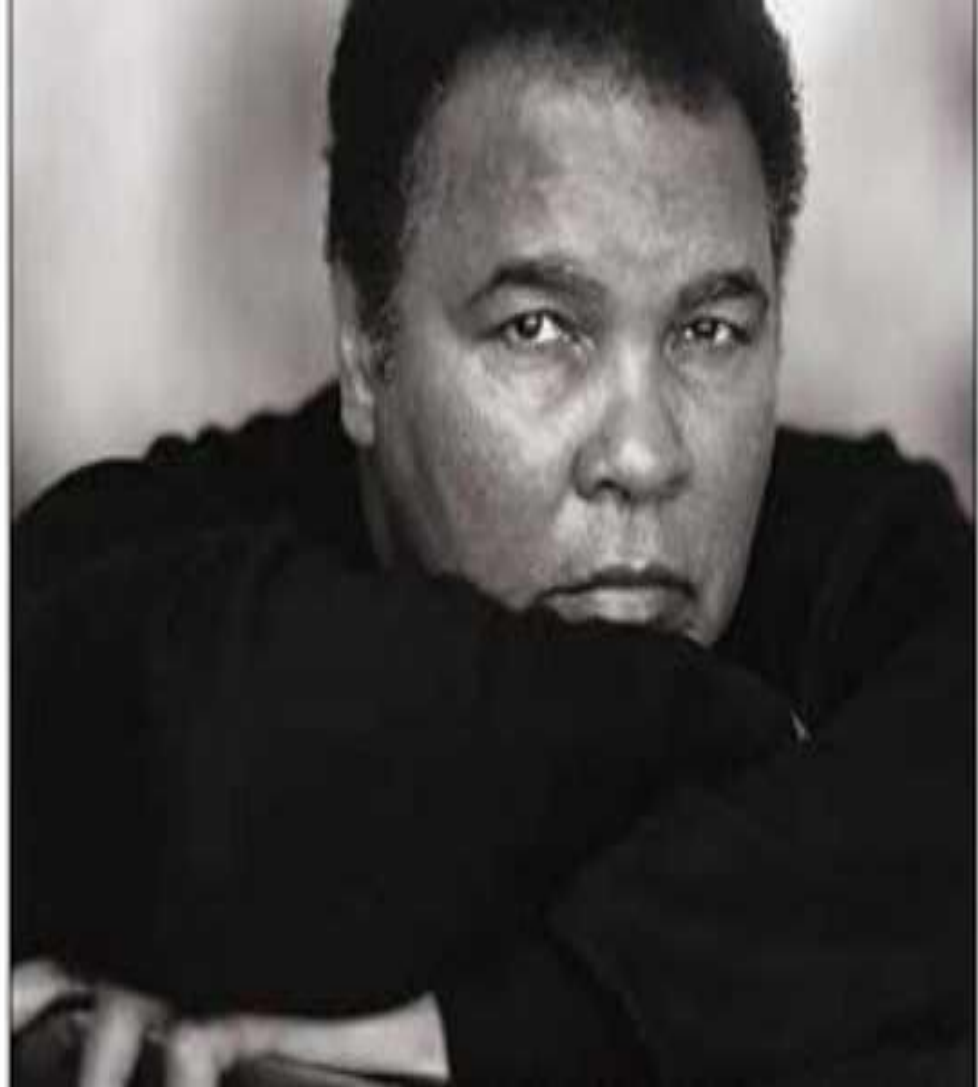
Outlines:

1. Introduction.
2. Definition of Parkinson's disease (PD).
3. Causes of Parkinson's disease (PD).
4. Clinical picture of Parkinson's disease (PD).
5. Diagnosis of Parkinson's disease (PD).
6. Treatment of Parkinson's disease (PD).
7. Nursing intervention of Parkinson's disease (PD).

Introduction

PD is the second most common neurodegenerative disorder after Alzheimer's disease and affects approximately seven million people globally and one million people in the United States. PD is more common in the elderly and rates rise from 1% in those over 60 years of age to 4% of the population over 80. The mean age of onset is around 60 years, although 5-10% of cases, classified as young onset PD, begin between the ages of 20 and 50. Some studies have proposed that it is more common in men than women, but others failed to detect any differences between the two sexes. The prevalence is between 8 and 18 per 100,000 person-years.

Some famous parkinsonian patients:



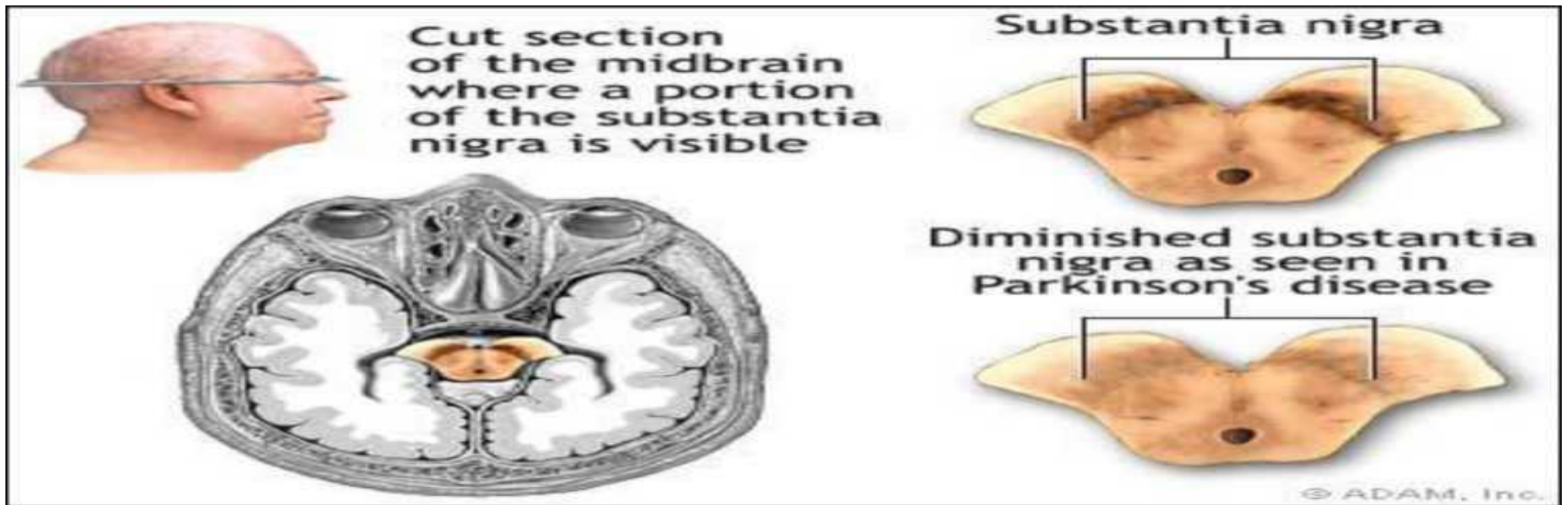
Adolf Hitler 's Muhammad Ali

Definition of Parkinson's disease (PD):

- Is a slowly progressive, degenerative disorder of the central nervous system that result from cell death in the region of the brain called substantia nigra and progressive loss of the nigrostriatal dopamenergic function which mainly affects the motor system. The symptoms generally come on slowly over time are resting tremors, bradykinesia, muscle rigidity and postural instability.

Causes:

- **Exact causes still yet unknown**; but may there is a risk factor such as; **gene mutation**, **toxins& trauma**.
- **PD results from deficiency of neurotransmitter dopamine.**
- **Neurons in the brain's substantia nigra degenerate, causing dopamine deficiency in the basal ganglia.**
- **As a result, an increased effect of acetylcholine causing symptoms of cholinergic excess such as rigidity, tremors, bradykinesia and postural instability.**



Signs and symptoms:

- Early symptoms: lack of facial expression and stiffening of the extremities.
- Cardinal features: includes the four **major symptoms**:
 - a) **Muscle rigidity**: Stiffens when the arm, leg or neck is moved **Hypomimia**: open eyes, lack of facial expression "mask like face", open mouth "drooling", Shuffling gait "cogwheel effect".

b) **Resting tremors:** tremor is most prominent at rest, it start first at one hand and arm then transferred to head, tremor started first unilateral then became bilateral & there is a pill rolling motion of the thumb against the fingers

c) **Bradykinesia:** slowness in movement result in weakness and reduced spontaneous movement, freezing during voluntary motor activity such as "slow steps of walking, difficult raising from chair, turning in bed, lack of arm swing during walking, smaller hand writing "micrographia", and change pattern of speech.

d) **Postural instability:** loss of postural reflexes, poor balance, shuffling gait, and head bent forwarded body move as one unit.

d) **Other symptoms:** constipation, urinary incontinence, decreases sexual function, slow monotonous speech, depression and dementia.

Diagnosis of Parkinson's disease (PD):

- No definitive tests for PD. PET scans can aid to determine levels of dopamine.
- Difficult to diagnose, many symptoms shared with other disorders.
- Medical history and neurological tests are conducted to diagnose.
- Usually, if two of the cardinal symptoms are present.

Treatment - Parkinson's Disease:

<i>Drug class</i>	<i>Mechanism of action</i>	<i>Side effect</i>
1- Dopamenergic "Levodopa, carbidopa"	Activate dopamine receptors	Nausea ,vomiting, hypotension and mental changes
2- Dopamine agonist	Activate dopamine receptors	Nausea, vomiting, hypotension and mental changes
3- Anticholinergics "Artane "	Block ACH receptors	Dry mouth, blurred vision and confusion.

Nursing intervention:

a) Assess patient present condition through:

1. Assess the length of time the symptoms appeared.
2. Assess the ability of the patient to carry out activities of daily living (ADLs) & assess the patient's mental condition.
3. Assess the patient's nutritional status, and monitor his weight.
4. Assess the patient's behavioral problems as presence of pain, constipation, hunger as well as any disruptive behavior.
5. Assess patient gait and posture
6. Assess tremor: it's important to distinguish between essential tremors and PD tremor; Parkinson tremor occurs on rest and also occurs with purposeful movement. "Essential" or "senile" tremor occurs only with action and disappears at rest.

b) Provide adequate nursing care through :

- 1) Provide emotional and psychological support for both patient and his family
- 2) Encourage the patient to use coping skills, help him to get involved in social activities to help in increasing self-esteem.
- 3) Provide adequate nutrition (small frequent and semisolid food).

- 4) A high protein diet can block the effect of Levodopa as well as alkaline products such as milk slow the absorption of drug.
- 5) Help the patient to establish regular bowel elimination, and be sure to use high elevated toilet seats for ease sitting.
- 6) Provide the patient adequate assistive devices.
- 7) Ask the physical therapist to develop a program of daily exercises which includes "stretching and postural exercises".
- 8) Protect the patient from injury by providing adequate safe environment to prevent falls.
- 9) Provide frequent warm bathes and massage to help relax muscles and relieve cramps.
- 10) Explain for the patient the importance of compliance with treatment regimen, and reporting the adverse effects of medication.
- 11) Provide appropriate referral if the patient speech affected to speech therapist as well occupational therapist.

Differentiation between senile tremor (Essential Tremor of Aging) and Parkinson's disease tremor

Type of tremor	Senile tremor	Parkinson's disease tremor
Action / Postural tremor	occurs during voluntary movement (e.g., holding a cup, writing) or when arms are held outstretched. (The tremor appeared with resistance)	Resting tremor – appears when muscles are relaxed Decreases with voluntary movement
Body parts affected:	Typically hands, head (yes-yes / no-no shake), voice Legs less commonly involved	Typically starts in one hand (“pill-rolling tremor”) May involve jaw or legs, rarely head or voice
Laterality	Usually bilateral and symmetric	Unilateral at onset, may become bilateral later
Other neurological signs	None – no rigidity, bradykinesia, or gait abnormalities	Present with rigidity, bradykinesia, shuffling gait, postural instability
Tremor frequency	High-frequency (6–12 Hz)	Low-frequency (4–6 Hz)

Thank You